PHYSICAL CHEMISTRY CUMULATIVE EXAM

March 4, 1995

Two of the most important experiments performed in the twentieth century were the Millikan "oil-drop" experiment performed in 1909 and that of Stern-Gerlach with the electron in 1921. Discuss what Millikan, Stern and Gerlach were trying to determine or prove with their experiments. Describe in detail, using diagrams when necessary, these experiments. Discuss the results of the experiments and their effects on chemistry and physics.

Physical Chemistry CUME
March 4, 2005

1. Give a sketch and the symmetry point group of the following molecules:

    1. SO_2 (bent)
    2. N_2O (linear)
    3. Naphthalene $C_{10}H_8$ (two fused benzene rings)
    4. *Cis*-dichlorotetrafluoroferrate(III), $[FeCl_2F_4]^{3-}$
    5. *Facial*-trichlorotrifluorocobaltate(III), $[CoCl_3F_3]^{3-}$
    6. PF_5 (the fluorines are in axial position)
    7. B_2H_6 (it is a bridged structure with no direct B-B bond, two hydrogens form bridges)
    8. NF_3 (pyramidal)
    9. Benzene
    10. Hydrogen peroxide (the stable conformation has an angle between the H-O-O' and O-O'-H' planes of about 110°)
    11. The white phosphorus molecule, P_4
    12. Ethylene (ethene)
    13. Acetylene (ethyne)
    14. CO_2
    15. Carbon tetrachloride